CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Match the **LIST-I** with **LIST-II**

	LIST-I (Redox Reaction)		**LIST-II (Type of Redox Reaction)**
A.	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	I.	Disproportionation reaction
B.	$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$	II.	Combination reaction
C.	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2V_{(s)} + 5CaO_{(s)}$	III.	Decomposition reaction
D.	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2H_2O_{(l)} + O_{2(g)}$	IV.	Displacement reaction

Choose the ***correct*** answer from the options given below:

(1) A-II, B-III, C-I, D-IV (2) A-IV, B-I, C-II, D-III

(3) A-III, B-IV, C-I, D-II (4) A-II, B-III, C-IV, D-I

Answer (4)

Sol. (A) $CH_4(g) + 2O_2(g) \xrightarrow{\Delta} 2CO_2(g) + H_2O(l) \Rightarrow$ combination reaction

(B) $2NaH(s) \xrightarrow{\Delta} 2Na(s) + H_2(g) \Rightarrow$ Decomposition reaction

(C) $V_2O_5(s) + 5Ca(s) \xrightarrow{\Delta} 2V(s) + 5CaO(s)$ $\Rightarrow$ Displacement reaction

(D) $2H_2O_2$ (aq.) $\xrightarrow{\Delta} 2H_2O(l) + O_2(g)$ $\Rightarrow$ Disproportionation reaction

52. Given below are two statements:

Statement I: D-glucose pentaacetate reacts with 2, 4-dinitrophenylhydrazine

Statement II: Starch, on heating with concentrated sulfuric acid at 100°C and 2-3 atmosphere pressure produces glucose.

In the light of the above statements, choose the ***correct*** answer from the options given below

(1) Both Statement I and Statement II are true

(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

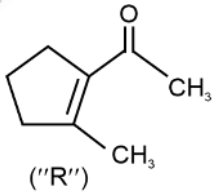
(4) Both Statement I and Statement II are false

Answer (3)

Sol. • D-glucose pentaacetate do not react with 2.4-DNP

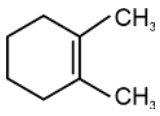
• Starch $\xrightarrow[\substack{2-3 \text{ atm} \\ 100°C}]{\text{conc.}H_2SO_4,\ \Delta}$ Glucose

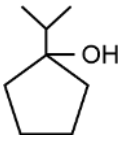
53. A molecule (''P'') on treatment with acid undergoes rearrangement and gives (''Q''). (''Q'') on ozonolysis followed by reflux under alkaline condition gives (''R''). The structure of (''R'') is given below.

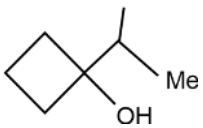


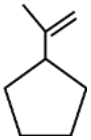
(''R'')

The structure of (''P'') is

(1) 

(2) 

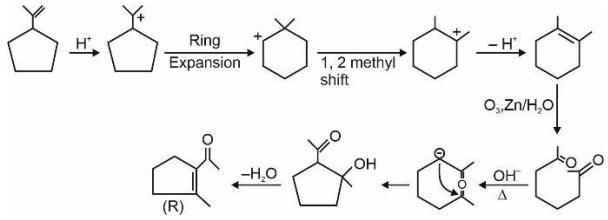
(3) 

(4) 

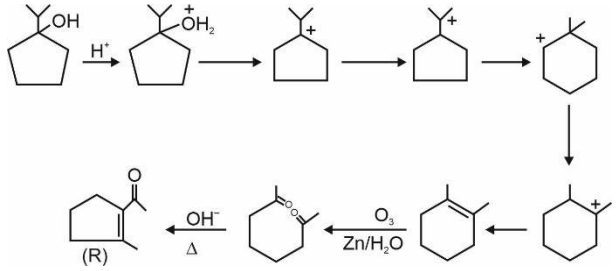


Answer (2 and 4)

Sol.



Or



54.

$[A]_0$ / mol L^{-1}	$t_{1/2}$ / min
0.100	200
0.025	100

For a given reaction R → P, $t_{1/2}$ is related to $[A]_0$ as given in table.

Given: log 2 = 0.30

Which of the following is <u>true</u>?

A. The order of the reaction is ½.

B. If $[A]_0$ is 1M, then $t_{1/2}$ is $200\sqrt{10}$ min

C. The order of the reaction changes to 1 if the concentration of reactant changes from 0.100 M to 0.500 M.

D. $t_{1/2}$ is 800 min for $[A]_0$ = 1.6 M

Choose the ***correct*** answer from the options given below:

(1) A, B and D only (2) A and C only

(3) C and D only (4) A and B only

Answer (1)

Sol. $t_{\frac{1}{2}} \propto (C_0)^{1-\eta}$

$$\frac{t_1}{t_2} = \left(\frac{C_1}{C_2}\right)^{1-\eta}$$

$$\Rightarrow \frac{200}{100} = \left(\frac{0.100}{0.025}\right)^{1-\eta}$$

$$\Rightarrow 2 = (4)^{1-\eta}$$

$$(1-\eta) = \frac{1}{2}$$

$$\eta = \frac{1}{2}$$

For $\eta = \frac{1}{2}$

$$\frac{-dA}{dt} = k(A)^{\frac{1}{2}}$$

$$\int_{C_0}^{C} \frac{dA}{(A)^{\frac{1}{2}}} = -\int_0^t kdt$$

$$\Rightarrow 2A^{\frac{1}{2}} = -kt$$

$$\Rightarrow \sqrt{C} - \sqrt{C_0} = \frac{-kt}{2}$$

$$\sqrt{C} - \sqrt{C_0} - \frac{kt}{2}$$

For $C_0 = 0.1 \Rightarrow t_{\frac{1}{2}} = 200$ min

$$\sqrt{\frac{c_0}{2}} = \sqrt{c_0} - \frac{kt}{2}$$

$$\frac{kt}{2} = \sqrt{c_0} - \sqrt{\frac{c_0}{2}}$$

$$\frac{kt}{2} = \sqrt{c_0}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$t_{\frac{1}{2}} = \frac{2\sqrt{c_0}}{k}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$200 = \frac{2\sqrt{0.1}}{k}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$



$$k = \frac{\sqrt{0.1}}{100}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

For $C_0 = 1M$

$$t_{\frac{1}{2}} = \frac{2\times100\left(\sqrt{2}\right)}{\sqrt{0.1}\left(\sqrt{2}-1\right)}\times\frac{\left(\sqrt{2}-1\right)}{\sqrt{2}}$$

$\Rightarrow 200\sqrt{10}\,\text{min}.$

$\Rightarrow$ B is correct

C is incorrect

For $C_0 = 1.6$ M

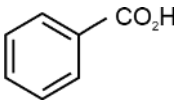
$$t_{\frac{1}{2}} = \frac{2\sqrt{1.6}\left(\sqrt{2}\right)\left(\sqrt{2}-1\right)\times100}{\sqrt{0.1}\left(\sqrt{2}-1\right)\left(\sqrt{2}\right)}$$

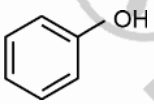
$$t_{\frac{1}{2}} = 400\times2\ \text{min}$$

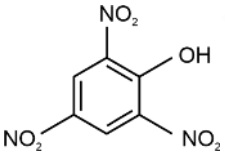
$$t_{\frac{1}{2}} = 800\ \text{min}$$

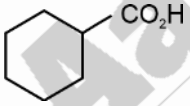
A, B and D are correct

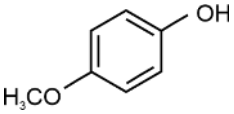
55. The compounds that produce CO_2 with aqueous $NaHCO_3$ solution are:

A. 

B. 

C. 

D. 

E. 

Choose the ***correct*** answer from the options given below:

(1) A, C and D only (2) A and C only

(3) A and B only (4) A, B and E only

Answer (1)

Sol. A, C, D produces CO_2 with $NaHCO_3$ as these are stronger acid as compared to H_2CO_3 (carbonic acid)

56. The incorrect decreasing order of atomic radii is

(1) Si > P > Cl > F (2) Mg > Al > C > O

(3) Be > Mg > Al > Si (4) Al > B > N > F

Answer (3)

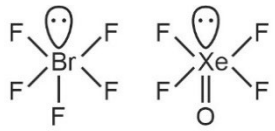
Sol. Correct order or atomic radii Be < Mg > Al > Si.

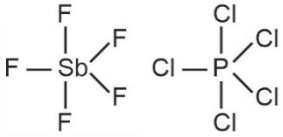
57. The molecules having square pyramidal geometry are

(1) BrF_5 & PCl_5 (2) BrF_5 & $XeOF_4$

(3) SbF_5 & $XeOF_4$ (4) SbF_5 & PCl_5

Answer (2)

Sol.  $\Rightarrow$ Square pyramidal

 $\Rightarrow$ Triangular bipyramidal

58. Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15K. If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following

(1) Volume of system increases

(2) The solid phase (ice) disappears completely

(3) Liquid phase disappears completely

(4) The amount of ice decreases

Answer (2)

Sol. $H_2O(s) \rightleftharpoons H_2O(l)$

On increasing pressure to 2 atm, freezing point of ice will decrease and solid phase (ice) will disappear completely.

59. In a multielectron atom, which of the following orbitals described by three quantum numbers will have same energy in absence of electric and magnetic fields?

A. $n = 1, l = 0, m_1 = 0$

B. $n = 2, l = 0, m_1 = 0$

C. $n = 2, l = 1, m_1 = 1$

D. $n = 3, l = 2, m_1 = 1$

E. $n = 3, l = 2, m_1 = 0$



Choose the **correct** answer from the options given below:

(1) A and B only (2) C and D only

(3) B and C only (4) D and E only

Answer (4)

Sol. (n + l) must be same.

	(n + l)
A	1
B	2
C	3
D	5
E	5

(n + l) value for D and E is same.

60. Which of the following oxidation reactions are carried out by both $K_2Cr_2O_7$ and $KMnO_4$ in acidic medium?

A. $I^- \rightarrow I_2$ B. $S^{2-} \rightarrow S$

C. $Fe^{2+} \rightarrow Fe^{3+}$ D. $I^- \rightarrow IO_3^-$

E. $S_2O_3^{2-} \rightarrow SO_4^{2-}$

Choose the **correct** answer from the option given below.

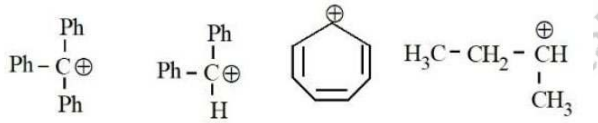
(1) A, B and C only (2) A, D and E only

(3) C, D and E only (4) B, C and D only

Answer (1)

Sol. I^- gets converted in I_2 by acidified $KMnO_4$.

61. The correct order of stability of following carbocations is :

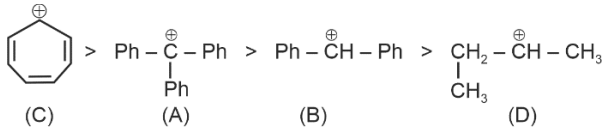


A B C D

(1) A > B > C > D (2) C > B > A > D

(3) C > A > B > D (4) B > C > A > D

Answer (3)

Sol. 

(C) (A) (B) (D)

C > A > B > D

62. Consider 'n' is the number of lone pair of electrons present in the equatorial position of the most stable structure of ClF_3. The ions from the following with 'n' number of unpaired electrons are

A. V^{3+} B. Ti^{3+}

C. Cu^{2+} D. Ni^{2+}

E. Ti^{2+}

Choose the ***correct*** answer from the options given below:

(1) A, D and E Only (2) B and C Only

(3) A and C Only (4) B and D Only

Answer (1)

Sol. n = 2

V^{3+} : $4s^0 3d^2$ (n = 2) $\Rightarrow$ A is correct

Ti^{3+} : $4s^0 3d^1$ (n = 1) $\Rightarrow$ B is not correct

Cu^{2+} : $4s^0 3d^9$ (n = 1) $\Rightarrow$ C is not correct

Ni^{2+} : $4s^0 3d^8$ (n = 2) $\Rightarrow$ D is correct

Ti^{2+} : $4s^0 3d^2$ (n = 2) $\Rightarrow$ E is correct

A, D and E are correct

Option (1) is correct

63. A weak acid HA has degree of dissociation x. Which option gives the correct expression of (pH – pK_a)?

(1) log(1 + 2x)

(2) $\log\left(\frac{1-x}{x}\right)$

(3) $\log\left(\frac{x}{1-x}\right)$

(4) 0

Answer (3)



Sol.

$$\begin{array}{ccccc} H & \rightleftharpoons & H^+ & + & A^- \\ C & & - & & - \\ C(1-x) & & Cx & & Cx \end{array}$$

$$K_a = \frac{[H^+](x)}{C(1-x)}$$

$$[H^+] = K_a \frac{(1-x)}{x}$$

$$\log H^+ = \log K_a + \log\left(\frac{1-x}{x}\right)$$

$$-\log H^+ = -\log K_a - \log\frac{1-x}{x}$$

$$pH = pK_a - \log\frac{1-x}{x}$$

$$pH - pK_a = \log\frac{x}{1-x}$$

64. The products A and B in the following reactions, respectively are

$$A \xleftarrow{Ag-NO_2} CH_3 - CH_2 - CH_2 - Br \xrightarrow{AgCN} B$$

(1) $CH_3-CH_2-CH_2-NO_2$, $CH_3-CH_2-CH_2-CN$

(2) $CH_3-CH_2-CH_2-ONO$, $CH_3-CH_2-CH_2-CN$

(3) $CH_3-CH_2-CH_2-ONO$, $CH_3-CH_2-CH_2-NC$

(4) $CH_3-CH_2-CH_2-NO_2$, $CH_3-CH_2-CH_2-NC$

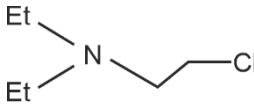
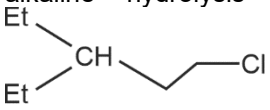
Answer (4)

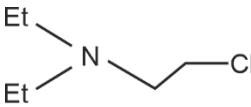
Sol. NO_2^- and CN^- are ambidentate nucleophiles

A : $CH_3-CH_2-CH_2-NO_2$

B : $CH_3-CH_2-CH_2-NC$

65. Given below are two statements:

Statement I:  will undergo alkaline hydrolysis at a faster rate than 

Statement II:  , intramolecular substitution takes place first by involving lone pair of electrons on nitrogen.

In the light of the above statements, choose the most appropriate answer from the options given below

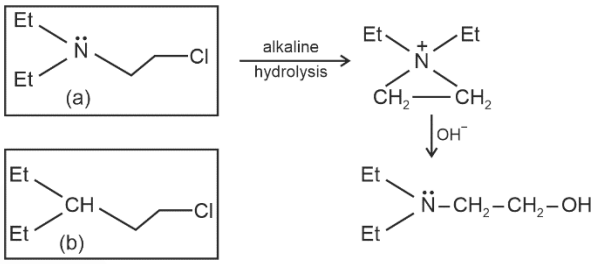
(1) Statement I is incorrect but Statement II is correct

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are incorrect

(4) Both Statement I and Statement II are correct

Answer (4)

Sol. 

Rate of (a) is faster than (b) because of its intramolecular substitution.

66. What is the freezing point depression constant of a solvent, 50 g of which contain 1 g non volatile solute (molar mass 256 g mol^{-1}) and the decrease in freezing point is 0.40 K?

(1) 1.86 K kg mol^{-1} (2) 5.12 K kg mol^{-1}

(3) 4.43 K kg mol^{-1} (4) 3.72 L kg mol^{-1}

Answer (2)

Sol. $\Delta T_f = 0.40$ K

$$(0.40) = (K_f)\left(\frac{1000}{(256)(50)}\right)$$

$$\boxed{(K_F) = 5.12 \text{ K kg mole}^{-1}}$$

67. The metal ion whose electronic configuration is not affected by the nature of the ligand and which gives a violet colour in non-luminous flame under hot condition in borax bead test is

(1) Ni^{2+} (2) Mn^{2+}

(3) Cr^{3+} (4) Ti^{3+}

Answer (1)



Sol. Ni^{2+} gives violet colour with borax bead test in non-luminous flame under hot conditions. Ni^{2+} has d^8 configuration which does not depend on nature of ligand present in octahedral field.

$Ni^{2+} \Rightarrow t_{2g}^6 eg^2$

68. Both acetaldehyde and acetone (individually) undergo which of the following reactions?

A. Iodoform Reaction

B. Cannizzaro Reaction

C. Aldol Condensation

D. Tollen's Test

E. Clemmensen Reduction

Choose the ***correct*** answer from the options given below:

(1) B, C and D only (2) A, B and D only

(3) A, C and E only (4) C and E only

Answer (3)

Sol.

	Name of Reaction	Acetaldehyde	Acetone
A.	Iodoform reaction	✓	✓
B.	Cannizaro reaction	×	×
C.	Aldol reaction	✓	✓
D.	Tollen's test	✓	×
E.	Clemmensen reduction	✓	✓

69. Given below are two statements :

Statement I : In the oxalic acid vs $KMnO_4$ (in the presence of dil H_2SO_4) titration the solution needs to be heated initially to 60°C, but no heating is required in Ferrous ammonium sulphate (FAS) vs $KMnO_4$ titration (in the presence of dill H_2SO_4)

Statement II : In oxalic acid vs $KMnO_4$ titration, the initial formation of $MnSO_4$ takes place at high temperature, which then acts as catalyst for further reaction. In the case of FAS vs $KMnO_4$, heating oxidizes Fe^{2+} into Fe^{3+} by oxygen of air and error may be introduced in the experiment.

In the light of the above statements, choose the ***correct*** answer from the options given below

(1) Both Statement I and Statement II are false

(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are true

Answer (4)

Sol. Heating is required in oxalic acid filtration due to High activation energy.

Heating is not required in FAS vs $KMnO_4$ titration because Fe^{2+} will get converted into Fe^{3+} by oxygen of air and error may be introduced in the experiment.

Both Statement-I and Statement-II are correct.

70. Consider the following elements In, Tl, Al, Pb, Sn and Ge.

The most stable oxidation states of elements with highest and lowest first ionisation enthalpies, respectively, are

(1) +2 and +3 (2) +4 and +3

(3) +4 and +1 (4) +1 and +4

Answer (2)

Sol. In, Tl, Al, Pb, Sn Ge

Lowest first I.E. = In $\Rightarrow$ most stable oxidation state $\Rightarrow$ +3

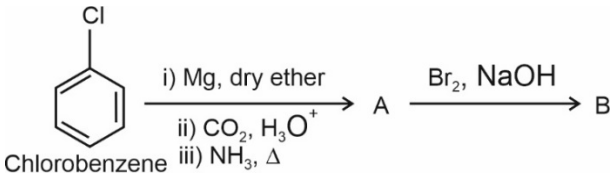
Highest first I.E. = Ge $\Rightarrow$ most stable oxidation state $\Rightarrow$ +4

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.



71. Consider the following sequence of reactions:



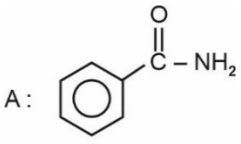
11.25 mg of chlorobenzene will produce ----- × 10^{-1} mg of product B.

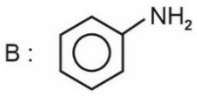
(Consider the reactions result in complete conversion.)

[Given molar mass of C, H, O, N and Cl as 12, 1, 16, 14 and 35.5 g mol^{-1} respectively]

Answer (93)

Sol.





moles of chlorobenzene = $\frac{11.25\times10^{-3}}{112.5}$

$= 10^{-4}$

moles of product B = 10^{-4}

mass of product B = $(10^{-4}) \times 93$

$= 93 \times 10^{-1}$ milligram

72. The formation enthalpies, ΔH_f° for $H_{(g)}$ and $O_{(g)}$ are 220.0 and 250.0 kJ mol^{-1}, respectively, at 298.15 K, and ΔH_f° for $H_2O_{(g)}$ is –242.0 kJ mol^{-1} at the same temperature. The average bound enthalpy of the O–H bond in water at 298.15 K is _____ kJ mol^{-1} (nearest integer).

Answer (466)

Sol. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$

(–242) = 440 + 250 – 2x

2x = 440 + 250 + 242

x = 466

73. Quantitative analysis of an organic compound (X) shows following % composition.

C : 14.5% Cl : 64.46%

H : 1.8%

(Empirical formula mass of the compound (X) is _____ × 10^{-1}

(Given molar mass in g mol^{-1} of C : 12, H : 1, O : 16, Cl : 35.5)

Answer (1655)

Sol. $C_{\left(\frac{14.5}{12}\right)}\ H_{\left(\frac{1.8}{1}\right)}\ Cl_{\left(\frac{64.46}{35.5}\right)}\ O_{\left(\frac{19.24}{16}\right)}$

$C_{1.21}\ H_{1.8}\ Cl_{1.81}\ O_{(1.2)}$

$\Rightarrow C_1\ H_{1.5}\ Cl_{1.5}\ O_1$

$\Rightarrow C_2H_3Cl_3O_2$

Empirical formula mass = $165.5\frac{gm}{mole}$

$= 1655 \times 10^{-1}$

74. The molarity of a 70% (mass/mass) aqueous solution of a monobasic acid (X) is _____ × 10^{-1} M(Nearest integer)

[Given : Density of aqueous solution of (X) is 1.25 g mL^{-1}

Molar mass of the acid is 70 g mol^{-1}]

Answer (125)

Sol. Moles of solute = $\frac{70}{70} = 1$

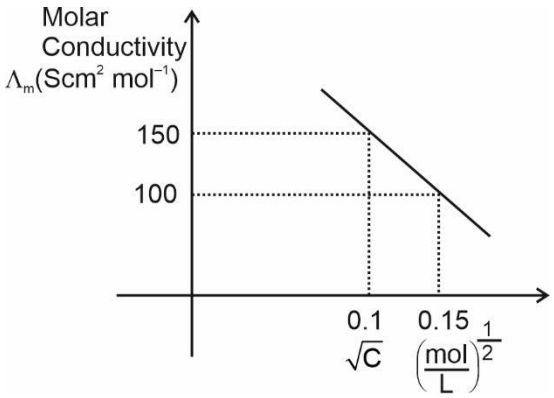
Volume of solution = $\frac{100}{1.25} = 80$ mL

$M = \frac{1}{80}\times1000 = 12.5$

$M = 125 \times 10^{-1}$



75. Given below is the plot of the molar conductivity vs $\sqrt{\text{concentration}}$ for KCl in aqueous solution.



If, for the higher concentration of KCl solution, the resistance of the conductivity cell is 100 W, then the resistance of the same cell with the dilute solution is 'x' W

The value of x is _____ (Nearest integer)

Answer (150)

Sol. $100 = \rho\left(\frac{l}{A}\right)$

For $\sqrt{C}$ = 0.15

l_m = 100

$$100 = \frac{k \times 1000}{0.15 \times 0.15}$$

$$k = \frac{15 \times 15}{10^5} = \frac{225}{105}$$

$$\rho = \frac{10^5}{225}$$

$$100 = \frac{10^5}{225}\left(\frac{l}{A}\right)$$

$$\frac{l}{A} = \frac{225}{1000}$$

For $\sqrt{C}$ = 0.1

l_m = 150

$$150 = \frac{k \times 1000}{10^{-2}}$$

$k = 15 \times 10^{-4}$

$$\rho = \frac{10^4}{15}$$

$$R = \rho\left(\frac{l}{A}\right)$$

$$R = \frac{10^4}{15} \times \frac{225}{1000}$$

$R \Rightarrow 150$ W

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